DMA Controller

Intel 8257

Direct Memory Access

A hardware device that allows I/O devices to directly access memory with less participation of the microprocessor

Q. Write Short notes on DMA Controller

Answer must include diagrams

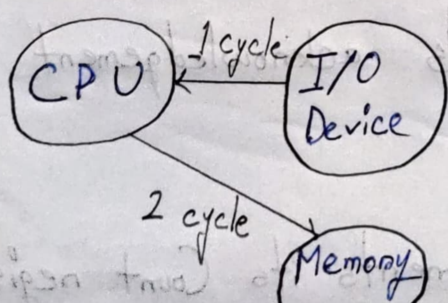
2 cycles to 1 cycle

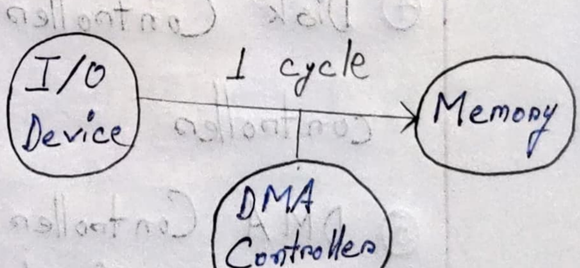
Q. How does DMAC accomplish data transfer in 1 cycle instead of 2 cycles

or

How does DMAC accomplish faster data transfer

Does NOT need to fetch/decode instruction only execute





See how DMAC works in YT

DMA is a process in which an external device takes over the control of system bus from CPU

Working Principle of DMA Controller

① CPU programs the DMA Controller by initializing its internal Address, Count & Control signal registers

② CPU instructs Disk Controller to read data from peripheral into its Buffer [DMA controller cannot function unless there is data in the Disk Controller's Buffer]

③ DMA Controller requests Disk Controller to transfer data from its Buffer to the Memory

④ Disk Controller sends acknowledgement to DMA controller

⑤ DMA Controller decrements its Count register and checks if it is zero

⑥ If Count != 0, steps ③-⑤ are repeated

If Count == 0, DMA controller sends an interrupt to the CPU

Q. What are the features of 8257

8257 Features: [Memorize]

1 ● Four channels which can be used over four I/O devices

○ Generates MARK signal to the peripheral device that 128 bytes have been transferred

5 ● Requires a single phase clock

○ Frequency ranges from 250 Hz to 3MHz

Do not miss important points (1-5)

2 ● Operates in 2 modes

Master mode

Slave mode

3 ● Has Priority Resolver

4 ● Has two 8-bit registers called mode set register and status register

Q. What are the modes of operations of 8257

8257 Modes of Operations:

① DMA Write

Data to be transferred from peripheral to memory

② DMA Read

Data to be transferred from memory to peripheral

③ DMA Verify

- No memory or I/O read/write control signals will be generated
- 8257 gains control of the system bus and will acknowledge the peripheral's DMA request for each DMA cycle
- Peripheral device can use these acknowledge signals to enable an internal access of each byte of a data block to execute some verification procedure

Q. What are the 8257 channels

8257 Channels:

- Four DMA channels (CH0- CH3)
- Each channel has two registers
  - > Address Register (16 bit)
  - > Terminal Count Register (2 bit DMA Operation + 14 bit data = 16 bit)
- Each channel can transfer data upto 64KB
- Each channel can be programmed independently
- Each channel can perform read, write and verify operations

8257 Pin Configuration:

- DRQ0 - DRQ3
  - > Individual channel request inputs used by the peripheral to obtain DMA cycles
  - > By default, DRQ0 has highest priority and DRQ3 has lowest (If not in rotating priority)

Q. How are requests & acknowledgements handled by DMA

Answer: DRQ 0- DRQ 3 & DACK 0 - DACK 3

o DACK0- DACK3

DMA Acknowledge output informs the peripheral connected to that channel that it has been selected for a DMA cycle

o D0- D7

> Bi-directional data bus lines

> In Slave Mode, 8 bits of data for DMA Address Register, Terminal Count Register or Mode Set Register is received through these lines

> When CPU reads a DMA Address register

>

Q. How does the terminal count register function?

Registers

① Terminal Count Register 16 bit

B_{15}	B_{14}														

0 0 ⇨ Verify transfer
0 1 ⇨ Write transfer
1 0 ⇨ Read transfer
1 1 ⇨ Illegal

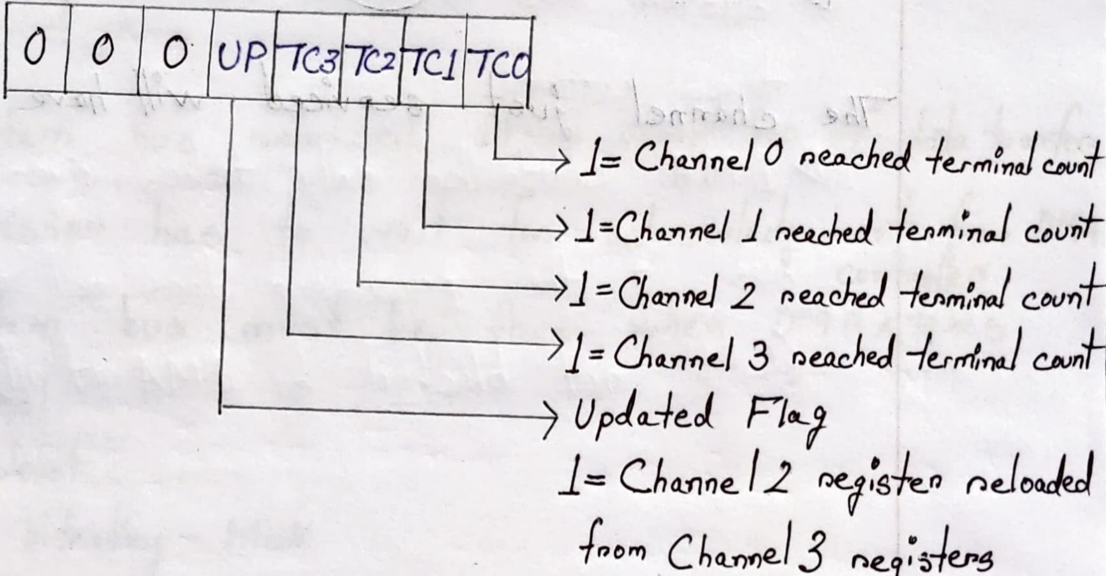
14-bit count

↳ Value in this 14 bits specifies the number of DMA cycles minus one

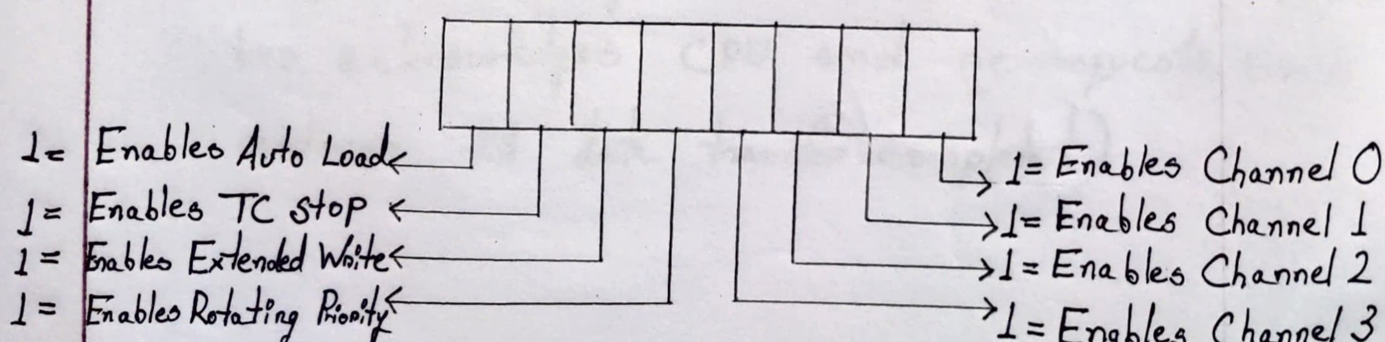
② Address Register

Q. How does the status register function?
Q. How does the mode set register function?
4 marks question

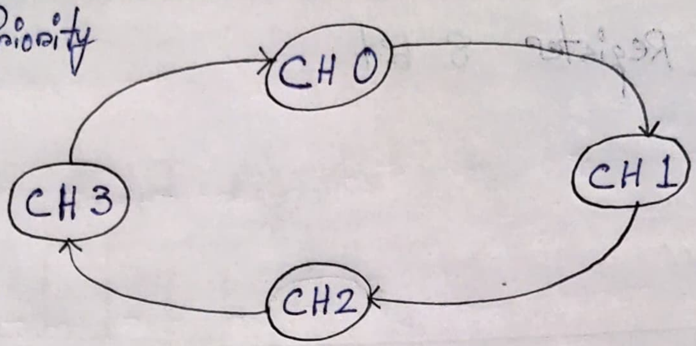
③ Status Register 8 bit



* ④ Mode Set Register 8 bit



Rotating Priority



The channel just serviced will have lowest priority

Auto Load

Allows Channel 2 to be used for block chaining operations

Extended Write

Duration of both MEMW and I/OW signals are extended

TC Stop

A channel is disabled after terminal count reaches its count

Q. How DMA controller steals clock cycles of processor/Why is Cycle Stealing Mode named "Clock Stealing"

Q. What are the types of DMA Mode transfer/Comparison between DMA Mode transfer

Types of Mode Transfer: Modes of Data Transfer

① Burst Mode

- Entire data or burst of block of data transferred at a time
- System bus released after completion of data transfer. Processor has to wait for acknowledgement from DMA controller
- System bus must be free when DMA controller takes control
- Quickest

② Cycle Stealing Mode

- Data transferred byte by byte / bytewise
- DMA controller requests processor to release system bus for a short time
- After each byte, DMA controller releases system bus, acknowledges CPU and re-requests (Loop continues till data transfer completed)

③ Transparent Mode

▷ Data transferred byte by byte / bytewise

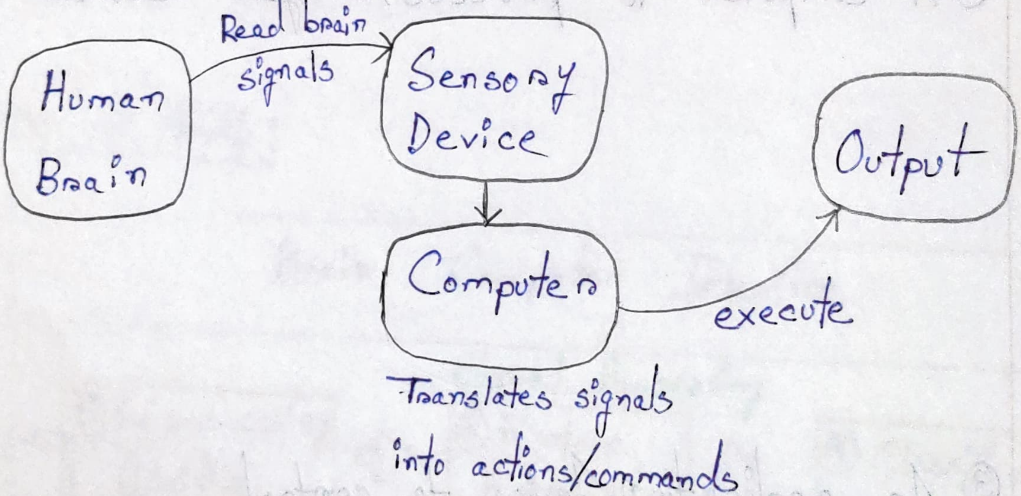
▷ System bus must be free when DMA controller takes control

▷ DMA controller only takes over the system bus if the processor does not require the bus (CPU not blocked by DMA at all)

Q1- Write Short Notes on BCI (Include figure)

Brain Computer Interfacing (BCI)

Definition: A direct connection between computer and human brain. where brain waves are read using sensors and these signals are translated into actions and commands that can control the computer.



Q2- What are the components of BCI?

Components of BCI:

① A device to measure brain activity

② A computer to process/analyze recorded activity

③ An application/device to control

④ Feedback

Feedback

Q3- Working Principle of BCI (figure)

Q4- What are the steps of BCI (figure)

Working Principle of BCI

② Process

Brain activity processed using a computer to interpret user's desired action

③ Control

A signal sent to application to execute desired command

④ Feedback

Feedback provided to user to indicate their action was successful

① Record

Brain Activity is recorded using special sensors

Steps of BCI:

Brain Computer Interface

Signal Processing

① Signal Acquisition

② Preprocessing (Digitize)

Digitizing analog values

③ Feature Extraction

Choosing Required Columns Only

④ Classification

Dividing data into categories

⑤ Signal Output

BCI Applications

feedback to user

User

Q7- Types of BCI / Difference between types of BCI

Applications of BCI → No questions in Exam

Types of BCI:

① Invasive BCI EEG

- Sensory device placed inside human brain grey matter
- Most accurate reading of brain waves

② Partially-Invasive BCI ECOG

- Sensory device placed inside scalp

③ Non-Invasive BCI implant

- Sensory device placed on head